

Recurrent Neural Networks (RNN)

RNN

is a deep learning architecture designed for **sequential data**. Processes data step-by-step while keeping information from previous steps using a hidden state.

Sequential Data:

- Text
- Speech
- Time Series
- Sensor Data
- Video Frames

Characteristic

- Captures temporal/sequential relationships
 - Uses memory from previous inputs
 - Same weights reused across time steps
 - Output depends on current input + previous hidden state
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Types of RNN Architectures

- **One-to-One**
 - Single input → single output
 - Image classification
 - **One-to-Many**
 - Single input → sequence output
 - Image captioning
 - **Many-to-One**
 - Sequence input → single output
 - Sentiment analysis
 - **Many-to-Many**
 - Sequence input → sequence output
 - Machine translation
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Advantages

- Good for sequential patterns
- Handles variable-length input
- Shared parameters reduce complexity

3 issues

- Variable Size of Input & Output Neurons
 - Too Much Computation
 - No Parameter Sharing
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LSTM (Long Short-Term Memory)

LSTM is an improved type of RNN designed to solve the vanishing gradient problem and remember long-term dependencies.

Characteristic

- Handles sequential data
 - Keeps long-term memory
 - Uses gates to control information flow
 - Better than traditional RNN for long sequences
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Components

- Cell State
 - Long-term memory
 - Hidden State
 - Current output/state
 - Forget Gate
 - Removes unnecessary information.
 - Input Gate
 - Adds new information to memory.
 - Output Gate
 - Controls final output.
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Pros

- Learns long-term dependencies
- Reduces vanishing gradient problem
- More stable than RNN

Cons

- **More computation**
 - **Slower training**
 - **More parameters than RNN**
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Applications

- **Sentiment Analysis**
- **Machine Translation**
- **Speech Recognition**
- **Time Series Forecasting**